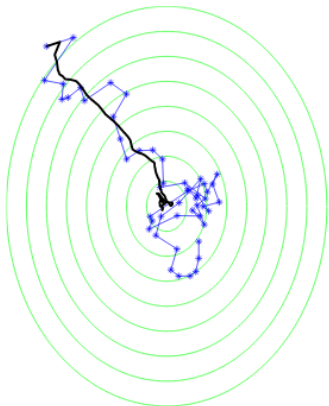
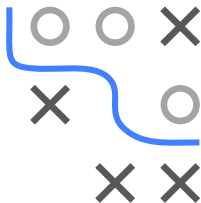


Optimization in Machine Learning

Advanced first order methods SGD



Learning goals

- Understand stochastic gradient descent
- Understand stochasticity and variance of SGD
- Understand effect of batch size and step size
- Understand stopping rules

STOCHASTIC GRADIENT DESCENT

- $g : \mathbb{R}^d \rightarrow \mathbb{R}$ objective, g is **average over functions**:

$$g(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n g_i(\mathbf{x}), \quad g \text{ and } g_i \text{ smooth}$$

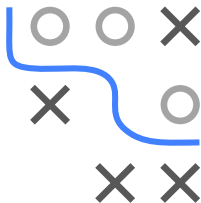
NB: We use g instead of f as objective, because f is used as model in ML

- Stochastic gradient descent (SGD) approximates the gradient

$$\begin{aligned} \nabla_{\mathbf{x}} g(\mathbf{x}) &= \frac{1}{n} \sum_{i=1}^n \nabla_{\mathbf{x}} g_i(\mathbf{x}) \quad := \quad \mathbf{d} \quad \text{by} \\ &\quad \frac{1}{|J|} \sum_{i \in J} \nabla_{\mathbf{x}} g_i(\mathbf{x}) \quad := \quad \hat{\mathbf{d}}, \end{aligned}$$

with random subset $J \subset \{1, 2, \dots, n\}$ called **mini-batch**

- This is done e.g. when computing the true gradient is **expensive**
- SGD is a **stochastic optimization** method as it uses stochastic gradient estimates



```

1: Initialize  $\mathbf{x}^{[0]}$ ,  $t = 0$ 
2: while stopping criterion not met do
3:   Randomly shuffle indices and partition into minibatches  $J_1, \dots, J_K$  of size  $m$ 
4:   for  $k \in \{1, \dots, K\}$  do
5:      $t \leftarrow t + 1$ 
6:     Compute gradient estimate with  $J_k$ :  $\hat{\mathbf{d}}^{[t]} \leftarrow \frac{1}{m} \sum_{i \in J_k} \nabla_{\mathbf{x}} g_i(\mathbf{x}^{[t-1]})$ 
7:     Apply update:  $\mathbf{x}^{[t]} \leftarrow \mathbf{x}^{[t-1]} - \alpha \cdot \hat{\mathbf{d}}^{[t]}$ 
8:   end for
9: end while

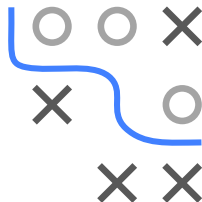
```

- Instead of drawing batches randomly we might want to go through the g_i sequentially (unless g_i are sorted in any way)
- Updates are computed faster, but also more stochastic:
 - In the simplest case, batch-size $m := |J_k|$ is set to $m = 1$
 - If n is a billion, computation of update is a billion times faster

- In ML, we perform ERM:

$$\mathcal{R}(\theta) = \frac{1}{n} \sum_{i=1}^n \underbrace{L(y^{(i)}, f(\mathbf{x}^{(i)} | \theta))}_{g_i(\theta)}$$

- for a data set $\mathcal{D} = ((\mathbf{x}^{(1)}, y^{(1)}), \dots, (\mathbf{x}^{(n)}, y^{(n)}))$
- a loss function $L(y, f(\mathbf{x}))$, e.g., L2 loss $L(y, f(\mathbf{x})) = (y - f(\mathbf{x}))^2$
- and a model class f , e.g., the linear model $f(\mathbf{x}^{(i)} | \theta) = \theta^\top \mathbf{x}$



- For large data sets, computing the exact gradient

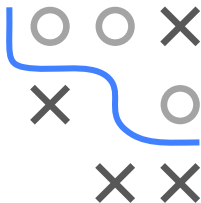
$$\mathbf{d} = \frac{1}{n} \sum_{i=1}^n \nabla_{\theta} L \left(y^{(i)}, f(\mathbf{x}^{(i)} | \theta) \right)$$

may be expensive or even infeasible to compute and is approximated by

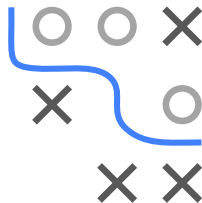
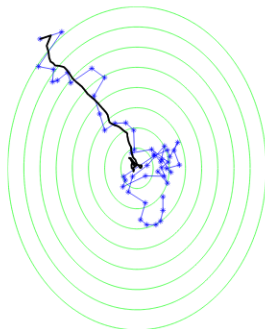
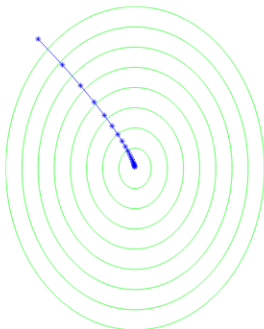
$$\hat{\mathbf{d}} = \frac{1}{m} \sum_{i \in J} \nabla_{\theta} L \left(y^{(i)}, f(\mathbf{x}^{(i)} | \theta) \right),$$

for $J \subset \{1, 2, \dots, n\}$ random subset

- **NB:** Often, maximum size of J technically limited by memory size



STOCHASTICITY OF SGD



► [Click for source](#)

Minimize $g(x_1, x_2) = 1.25(x_1 + 6)^2 + (x_2 - 8)^2$ **Left: GD, Right: SGD**

Black line shows average value across multiple runs

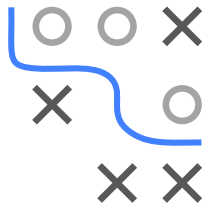
SGD trajectory much more noisy than of GD (due to stochastic gradient estimates)

STOCHASTICITY OF SGD

- Assume batch size $m = 1$ (statements also apply for larger batches)
- **(Possibly) suboptimal direction:** Approximate gradient $\hat{\mathbf{d}} = \nabla_{\mathbf{x}} g_i(\mathbf{x})$ might point in suboptimal (possibly not even a descent!) direction
- **Unbiased estimate:** If J drawn i.i.d., approximate gradient $\hat{\mathbf{d}}$ is an unbiased estimate of gradient $\mathbf{d} = \nabla_{\mathbf{x}} g(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n \nabla_{\mathbf{x}} g_i(\mathbf{x})$:

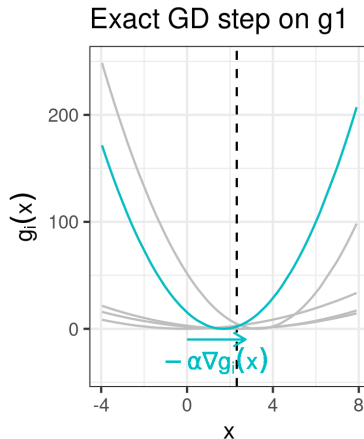
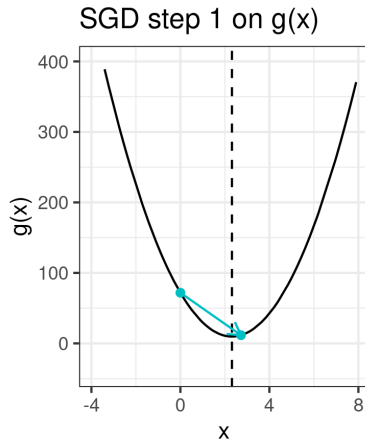
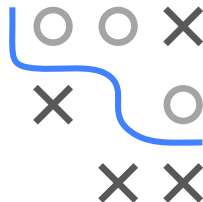
$$\begin{aligned}\mathbb{E}_i [\nabla_{\mathbf{x}} g_i(\mathbf{x})] &= \sum_{i=1}^n \nabla_{\mathbf{x}} g_i(\mathbf{x}) \cdot \mathbb{P}(i = i) \\ &= \sum_{i=1}^n \nabla_{\mathbf{x}} g_i(\mathbf{x}) \cdot \frac{1}{n} = \nabla_{\mathbf{x}} g(\mathbf{x}).\end{aligned}$$

- **Conclusion:** SGD might perform single suboptimal moves, but moves in “right direction” **on average**



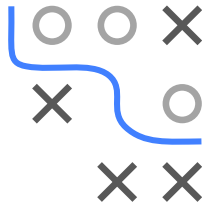
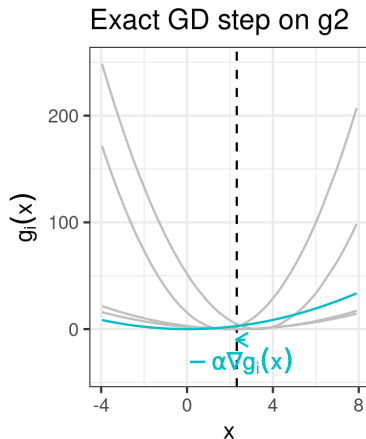
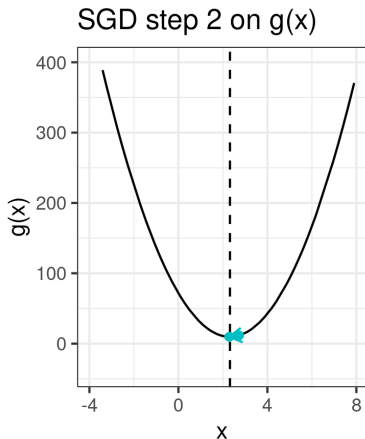
ERRATIC BEHAVIOR OF SGD

Example: $g(\mathbf{x}) = \sum_{i=1}^5 g_i(\mathbf{x})$, g_i quadratic, batch size $m = 1$
(Note that each SGD step for $m = 1$ corresponds to an exact GD step on one of the g_i .)

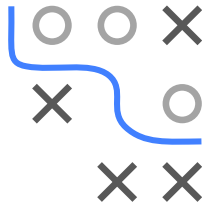
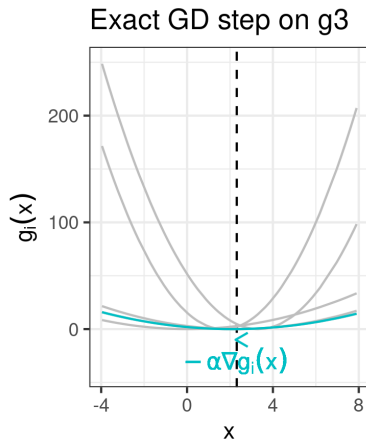


ERRATIC BEHAVIOR OF SGD

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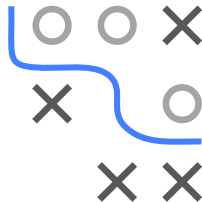
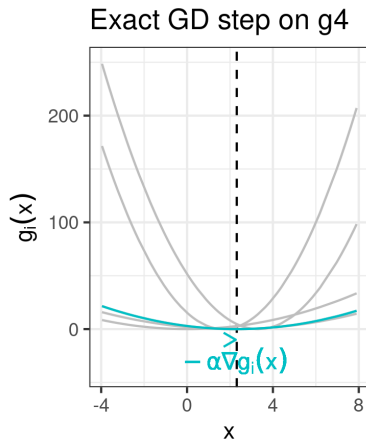
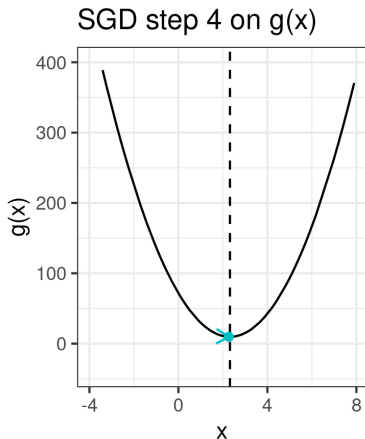


Example: $g(\mathbf{x}) = \sum_{i=1}^5 g_i(\mathbf{x})$, g_i quadratic, batch size $m = 1$



ERRATIC BEHAVIOR OF SGD

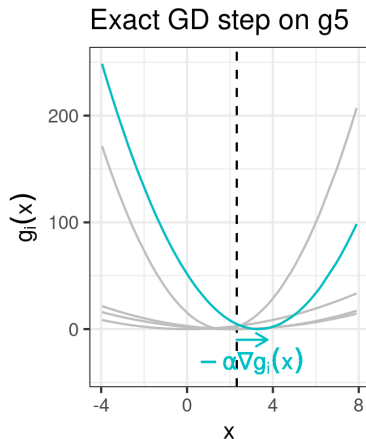
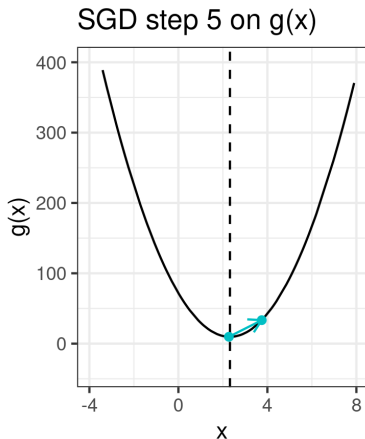
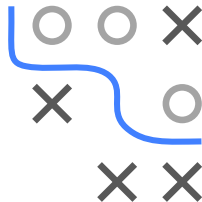
Example: $g(\mathbf{x}) = \sum_{i=1}^5 g_i(\mathbf{x})$, g_i quadratic, batch size $m = 1$



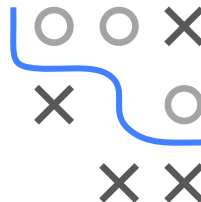
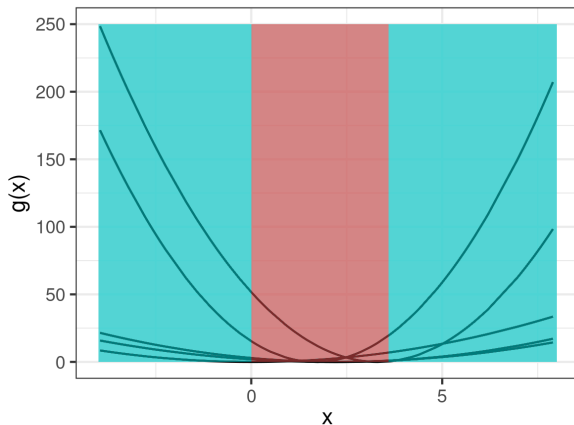
ERRATIC BEHAVIOR OF SGD

Example: $g(\mathbf{x}) = \sum_{i=1}^5 g_i(\mathbf{x})$, g_i quadratic, batch size $m = 1$

In iteration 5, SGD performs a suboptimal move away from the minimum:



ERRATIC BEHAVIOR OF SGD



Blue area: Each $-\nabla g_i(\mathbf{x})$ points towards minimum of $g(\mathbf{x})$

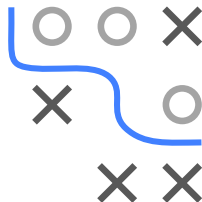
Red area (“confusion area”): $-\nabla g_i(\mathbf{x})$ might point away from minimum
and perform a suboptimal move

ERRATIC BEHAVIOR OF SGD

- At location $\mathbf{x}$, “confusion” is captured by **variance of gradients**

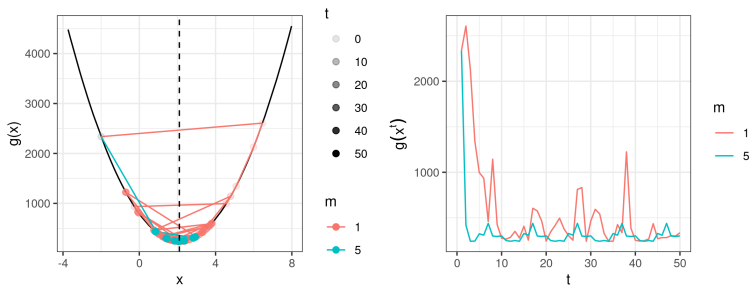
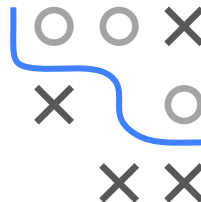
$$\frac{1}{n} \sum_{i=1}^n \|\nabla_{\mathbf{x}} g_i(\mathbf{x}) - \nabla_{\mathbf{x}} g(\mathbf{x})\|^2$$

- If term is 0, next step goes in gradient direction (for each i)
- If term is small, next step *likely* goes in gradient direction
- If term is large, next step likely goes in direction different than gradient
- Consequently, SGD has worse convergence properties than GD
- **But:** Can be controlled by tuning **batch size** and **step size**.



EFFECT OF BATCH SIZE

- The larger the batch size m
 - the better the approximation to $\nabla_x g(\mathbf{x})$
 - the lower the risk of performing steps in the wrong direction
 - the lower the variance
 - **But:** Maximum batch size is usually limited by computational resources (memory)



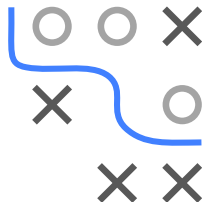
Example continued: Batch size $m = 1$ vs. $m = 5$

EFFECT OF STEP SIZE

- The smaller the step size α
 - the smaller erratic steps in potentially wrong directions
 - the lower the effect of gradient variance
 - **But:** Risk of impaired performance by not moving far enough
- SGD converges for sufficiently smooth functions if

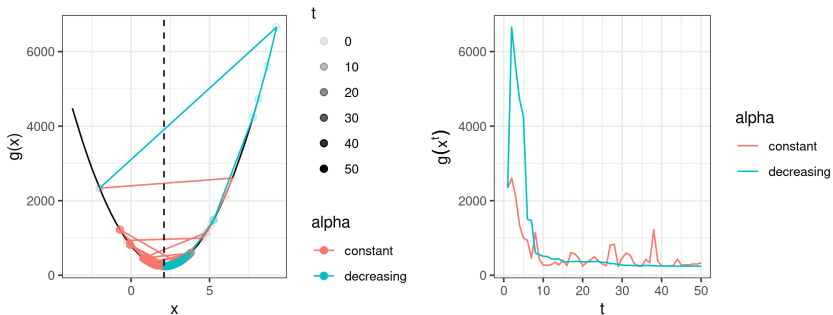
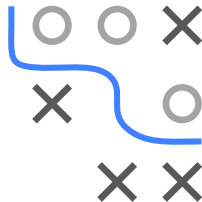
$$\frac{\sum_{t=1}^{\infty} (\alpha^{[t]})^2}{\sum_{t=1}^{\infty} \alpha^{[t]}} = 0$$

(“how much noise affects you” by “how far you can get”)



STEP SIZE CONTROL FOR SGD

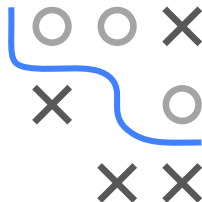
- Popular solution: Decreasing step size
 - Initially, step size shall be large enough to make progress
 - In later iterations, step size shall be small enough to reduce effect of gradient variance



Example continued: Step size $\alpha^{[t]} = 0.2/t$

Decreasing step size leads to more stable convergence

STEP SIZE CONTROL FOR SGD



Why not Armijo-based step size control?

- Backtracking line search or other approaches based on Armijo rule usually not suitable: Armijo condition

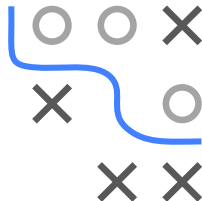
$$g(\mathbf{x} + \alpha \mathbf{d}) \leq g(\mathbf{x}) + \gamma_1 \alpha \nabla g(\mathbf{x})^\top \mathbf{d}$$

requires evaluating full gradient

- But SGD is used to *avoid* expensive gradient computations
- Research aims at finding inexact line search methods that provide better convergence behavior, e.g., [► Vaswani et al. 2019](#)

STOPPING RULES FOR SGD

- For GD: We usually stop when gradient is close to 0 (i.e., we are close to a stationary point)
- For SGD: Individual gradients do not necessarily go to zero, and we cannot access full gradient
- Practicable solution for ML:
 - Measure the validation set error after T iterations
 - Stop if validation set error is not improving



SGD AND ML

General remarks:

- SGD is a variant of GD
- SGD particularly suitable for large-scale ML when evaluating gradient is too expensive
- Thus, SGD and variants are the most commonly used methods in modern ML with e.g. neural networks
- Note that even for the linear model and quadratic loss, where a closed form solution is available, SGD is often used if the size n of the dataset is too large and the design matrix does not fit into memory

